

JACOB COOK

Detection Engineer | SOC Analyst | Wazuh · Azure AD · Okta · Python · SOAR

Sacramento, CA · (916) 314-5358 · jacobcookofficial@gmail.com

[linkedin.com/in/jcook-dev](https://www.linkedin.com/in/jcook-dev) · github.com/jacobdcook · jacobdcook.com

Detection-focused security professional with hands-on experience building SIEM environments, engineering detection rules, and automating incident response workflows. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX) with a BS in Computer Science and an MS in Cybersecurity in progress. Six years of operational systems experience in monitoring, root cause analysis, and incident coordination that translates directly into SOC intuition. Building in Wazuh, Okta, Azure, and Python daily.

TECHNICAL SKILLS

Detection & SOC: Threat Detection, SIEM (Wazuh), Alert Triage, Incident Documentation, Phishing Analysis, IOC Analysis, Playbook Automation, SOAR

Cloud & Identity: Azure AD, Okta, SSO / RBAC, Terraform, Microsoft Sentinel (familiar)

Systems: Windows Endpoints, Linux (Ubuntu / CentOS), Wireshark, Networking, System Hardening, VirtualBox

Development: Python, Bash, SQL, FastAPI, React, Git

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code

2026

- Authored 6 Sigma-format detection rules for Intune MDM abuse mapped to MITRE ATT&CK, with enrichment logic that eliminated 100% of false positives in a 100-event simulation (80 benign suppressed, 20 malicious caught) using KV store-style admin baseline lookups.
- Documented 3 benign false-positive scenarios per rule (18 total), covering VPN logins, scheduled maintenance, and PIM activations so analysts can triage confidently without escalating known-good activity.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py), reproducible against synthetic data, not a live tenant.

Blue Team SOC Monitoring Lab: Wazuh SIEM

2026

- Built a full Wazuh SIEM stack from scratch and validated 12 detection rule patterns against 242 simulated auth-style events, separating true-positive threats from benign noise with 100% precision after tuning.
- Reduced false-positive alert volume by 100% by tuning out all 79 benign-only matches (cron jobs, service accounts, scheduled deployments) without dropping a single true positive.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py). All metrics reproducible.

SOAR-lite: Incident Response Orchestrator

2026

- Eliminated 4 manual Tier 1 steps (classification, IOC enrichment, notification, ticket creation) by building automated SIEM-triggered playbooks, reducing triage time by over 60% in lab testing.
- Built end-to-end alert automation: webhook ingestion from Wazuh, automatic playbook routing by event type, IP blocking, Slack notification, and incident documentation with zero analyst intervention required.
- Delivered a working Tier 1 to Tier 2 handoff system that closes the gap between raw detection and documented, actionable response.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py). Tests run against the real FastAPI app, not mocked endpoints.

Phishing Analysis Lab

2026

- Triaged 5 real-world phishing samples end-to-end: header inspection, SPF/DKIM/DMARC validation, URL defanging, VirusTotal enrichment, IOC extraction, and attacker infrastructure mapping, achieving 100% accurate threat classification against known IOC ground truth.
- Delivered Tier 1 quality triage reports with documented IOCs, confidence assessments, and escalation decisions that mirror production SOC analyst output.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6)

2024

- Secured a production AI platform for corporate stakeholders by engineering RBAC and Azure AD SSO, enforcing data privacy requirements that enabled executive-level adoption.
- Shipped an enterprise-grade platform on deadline as part of a six-person team under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response

2019 – 2025

- Maintained 99.5% system uptime across automated sortation and conveyor infrastructure by monitoring sensors, alerts, and operational dashboards across rotating shifts, the same continuous monitoring discipline that defines SOC Tier 1 work.
- Resolved 200+ system anomalies over six years through structured root cause analysis, producing compliance-tracked documentation that reduced repeat incidents by an estimated 30% and supported audit readiness.
- Reduced mean time to recovery by an estimated 20% by coordinating cross-team incident response during outages, including escalation, triage prioritization, and post-incident reporting that directly mirrors SOC workflows.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)